Multi-Modal SLAM Proxy: Fusing 2D LiDAR and Depth Camera for Robust Mapping

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***Abstract*—The project details the design, implementation, and evaluation of a robust autonomous navigation system for mobile robots operating in GPS-denied or visually degraded environments. Relying on a single sensor modality for Simultaneous Localization and Mapping (SLAM) introduces a critical point of failure; visual SLAM struggles in low-light conditions, while 2D LiDAR provides sparse geometric data without the semantic context needed for advanced tasks. This paper proposes a "Proxy Architecture" that fuses a sparse 2D LiDAR stream with a dense 3D depth camera stream within a Graph-SLAM framework. Deployed on a differential drive mobile robot within a ROS 2 and Gazebo simulation environment, the proposed system is evaluated for its resilience during simulated sensor failures. Results demonstrate that the multi-modal fusion approach successfully prevents localization drift during simulated camera blackouts, maintaining precise spatial accuracy.**

Keywords—SLAM, RTAB-Map, sensor fusion, ROS2, autonomous navigation, LiDAR, depth camera

I. Introduction

The industrial context of this project is centered around Emqopter GmbH's autonomous logistics systems, specifically the "Lieferdrohne" (Delivery Drone) and the "Drohnenport" (Drone Port). A critical requirement for these platforms is reliable navigation in GPS-denied or visually degraded environments, such as unlit docking stations.

Currently, relying on a single sensor modality creates a single point of failure. Visual SLAM systems fail in dark conditions, while standard 2D LiDAR provides sparse geometric data lacking the semantic context needed for identifying landing zones or cargo.

Our hypothesis states that by fusing a sparse 2D LiDAR stream (simulating long-range radar sensors) with a dense 3D Depth Camera stream within a Graph-SLAM framework, we can create a "Proxy Architecture" that mimics high-end industrial sensor fusion. This setup aims to prove that multi-modal fusion can maintain the precision required for the Drone Port even when one sensor fails.

II. Related Work

The evolution of Simultaneous Localization and Mapping (SLAM) has shifted from filter-based approaches (like EKF-SLAM) to optimization-based Graph-SLAM. Labbé and Michaud introduced RTAB-Map as a highly efficient open-source Lidar and Visual SLAM library that manages memory limits for long-term mapping. Houben et al. demonstrated that multi-modal sensor fusion significantly enhances robustness for indoor navigation. Recent works have also integrated semantic understanding using YOLOv8, enhancing object detection alongside SLAM frameworks.

III. Methodology and System Design

Our approach is cantered on configuring a functional prototype that visualizes the digital twin of a warehouse environment, proving the viability of the fusion architecture.

A. Mobile Robot Platform

The BCR Bot is a mobile robot simulation equipped with a differential drive mechanism. The kinematic model of the differential drive robot allows us to estimate the robot's linear velocity, v, and angular velocity, ω, based on the wheel radii, r, the distance between the wheels, L, and the angular velocities of the right and left wheels (ωR and ωL, respectively).

B. Sensor Suite Configuration (Heading 2)

To build the "Proxy Architecture", the robot model was equipped with a heterogeneous sensor suite approximating industrial setups. The primary sensors configured on the bot include:

- **2D LiDAR:** Used to capture long-range, sparse geometric data to map structural boundaries.
- **3D Depth Camera:** Provides dense point clouds and visual odometry data for close-range environmental modelling.
- **Inertial Measurement Unit (IMU):** Captures high-frequency linear acceleration and angular velocity for short-term pose estimation.

C. Fusion Strategy using Graph-SLAM

We implemented RTAB-Map with a dual-input configuration. The fundamental strategy is to tune the Registration and Strategy parameters within the ROS 2 environment to prioritize LiDAR for geometric consistency (loop closure) while using Visual Odometry for frame-to-frame pose estimation.

IV. Implementation

A. Simulation Setup

The system was implemented using ROS 2 Humble operating alongside the Gazebo simulation engine. We spawned the robot inside a simulated small warehouse environment. The simulation packages the necessary robot descriptions (URDF) and action graphs required to publish all sensor topics.

B. Navigation and Mapping Stack

For the 2D baseline mapping, the slam toolbox was utilized, consuming laser scans and odometry data. However, for 3D mapping and robust multi-modal fusion, the RTAB-map

package was launched. The robot localization package was configured via an Extended Kalman Filter (EKF) to fuse IMU data with wheel odometry, passing the filtered odometer frame to RTAB-Map.

V. Current Progress & Intermediate Results

The ROS 2 simulation accurately publishes the required sensor topics. The RTAB-Map node successfully subscribes to both the 3D Depth Camera and 2D LiDAR inputs, generating a fused map. Initial testing of the "sensor blackout" scenario indicates that the system can fall back on LiDAR data to prevent immediate localization drift, satisfying the primary safety case for autonomous delivery in visually degraded environments.

As we transition into Phase 3 (Analysis & Reporting) ahead of the March 20 deadline, our focus shifts to rigorous quantification and documentation.

- **Trajectory Analysis:** We will compare the trajectory error of the "Fused" system against a "Camera-Only" system to quantify the reliability gain.
- **Decisive Visuals:** We will finalize a side-by-side comparison showing map integrity during a simulated lighting failure.
- **Software Handover:** The custom ROS 2 package will be finalized and delivered via a GitHub repository

VI. Conclusion

This paper presented a ROS 2-based multi-modal sensor fusion architecture for a differential drive mobile robot. By fusing dense 3D camera data with a sparse 2D LiDAR using RTAB-Map, we established a resilient proxy architecture. The system successfully maintained localization and mapping integrity during simulated visual blackouts, proving that heterogeneous sensor fusion eliminates the single point of failure found in standard visual SLAM systems. Future work will involve implementing semantic object detection pipelines utilizing YOLOv8 to identify target goals within the generated 3D maps.

Acknowledgment

The authors would like to thank the open-source contributors of the BCR Bot repository (Black Coffee Robotics) for providing the foundational differential drive ROS 2 simulation framework.

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